Measurement record, 2026-10

This record lists published comparisons from the benchmark boards on NVIDIA L40S (sm_89) and AMD Instinct MI300X (gfx942). All Anon measurements use the default *identical* configuration. The comparisons report ties and slower runtimes. Raw board measurements are preserved. The boards retain the Anon GPU arm and opponent measurements on the same GPU model and dataset. Each algorithm and dataset has one row per opponent; inference calls ("infer") follow the fit as rows of their own where recorded. The NVIDIA and AMD snapshots are the release 0.8.55 campaign boards pinned by results/measurement-sources.json. The records retain the release wheels, harness revisions, effective settings, input dimensions and paired opponent observations. Each successful observation of our implementation has one scored round. Warm-up, synchronization and timing boundaries are retained in the source records. Times are wall-clock medians in milliseconds. Whole-operation comparisons include host-to-device transfer; † marks kernel-only comparisons on both sides and § marks asymmetric transfer accounting. The table generator uses the clock operands recorded by each admitted pair's audit. Quality is the metric the board stores for each algorithm: held-out metrics for trees, and for the other families one float64 function over each arm's outputs. One metric is shown per row, named in the first quality column. The column "my ms ÷ their ms" divides my median by the opponent's and shows it as a multiple; below 1.00× my median was the lower. Source paths, hashes, dates and cell counts of the board files are in results/measurement-sources.json.

1 NVIDIA L40S

Table 1: NVIDIA L40S: fit and inference comparisons. Times in milliseconds; ratios divide my runtime by the opponent's. Quality is shown alongside runtime. Snapshot: October 30, 2026. § my time includes host-to-device transfer; the opponent's excludes it.

family	algorithm	dataset	my ms	opponent	their ms	my ms ÷ their ms	my quality	their quality
neural	gemm	gaussian	46.0	torch-compile-bf16 2.13.0+cu129	34.0	**1.35**×	max_rel_err_vs_fp64 2.4e−7	0.00375
			46.0	torch-compile-fp32 2.13.0+cu129	39.0	**1.18**×		1.4e−6
			46.0	torch-compile-tf32 2.13.0+cu129	35.6	**1.29**×		2.78e−4
			46.0	torch-eager-bf16 2.13.0+cu129	34.4	**1.34**×		0.00375
			46.0	torch-eager-fp32 2.13.0+cu129	36.4	**1.26**×		1.4e−6
			46.0	torch-eager-tf32 2.13.0+cu129	35.0	**1.31**×		2.78e−4
neural	gemm-bf16	gaussian	65.9	torch-compile-bf16 2.13.0+cu129	40.4	**1.63**×	max_rel_err_vs_fp64 1.15e−7	0.00276
			65.9	torch-eager-bf16 2.13.0+cu129	39.7	**1.66**×		0.00276
neural	gemm-int8	gaussian	43.8	torch-compile-int8 2.13.0+cu129	38.7	**1.13**×	max_rel_err_vs_fp64 0	0
			43.8	torch-eager-int8 2.13.0+cu129	37.7	**1.16**×		0
neural	lm-forward	bytes	44.9	torch-compile-bf16 2.13.0+cu129	31.1	**1.45**×	mean_nll 9.02	9.02
			44.9	torch-compile-fp32 2.13.0+cu129	35.6	**1.26**×		9.02
			44.9	torch-compile-tf32 2.13.0+cu129	34.0	**1.32**×		9.02

NVIDIA L40S, continued

family	algorithm	dataset	my ms	opponent	their ms	my ms ÷ their ms	my quality	their quality
			44.9	torch-eager-bf16 2.13.0+cu129	34.2	**1.31**×		9.02
			44.9	torch-eager-fp32 2.13.0+cu129	36.7	**1.22**×		9.02
			44.9	torch-eager-tf32 2.13.0+cu129	34.0	**1.32**×		9.02
			44.9	torch-graph-fp32 2.13.0+cu129	32.6	**1.38**×		9.02
			44.9	torch-graph-tf32 2.13.0+cu129	31.6	**1.42**×		9.02
neural	lm-train-step	bytes	35.5	torch-compile-bf16 2.13.0+cu129	10.8	**3.28**×	loss_last_step 8.41	8.40
			35.5	torch-compile-fp32 2.13.0+cu129	24.2	**1.47**×		8.41
			35.5	torch-compile-tf32 2.13.0+cu129	18.4	**1.93**×		8.41
			35.5	torch-eager-bf16 2.13.0+cu129	22.5	**1.58**×		8.41
			35.5	torch-eager-fp32 2.13.0+cu129	22.3	**1.59**×		8.41
			35.5	torch-eager-tf32 2.13.0+cu129	19.4	**1.83**×		8.41
neural	mamba1-forward	gaussian	3.43	mamba-ssm-fp32 2.3.2.post1+ge9594ce1c732	3.31	**1.04**×	rel_fro_vs_torch_eager_fp32 6.38e−9	5.04e−9
			3.43	mamba-ssm-tf32 2.3.2.post1+ge9594ce1c732	3.20	**1.07**×		6.4e−7
neural	mamba2-forward	gaussian	5.57	mamba-ssm-fp32 2.3.2.post1+ge9594ce1c732	3.28	**1.70**×	rel_fro_vs_torch_eager_fp32 1.5e−7	8.53e−8
			5.57	mamba-ssm-tf32 2.3.2.post1+ge9594ce1c732	2.81	**1.98**×		9.07e−5
neural	mamba3-forward	gaussian	3.95	mamba-ssm-fp32 2.3.2.post1+ge9594ce1c732	3.23	**1.22**×	rel_fro_vs_torch_eager_fp32 5.55e−8	1.98e−4
			3.95	mamba-ssm-tf32 2.3.2.post1+ge9594ce1c732	3.07	**1.29**×		1.99e−4
neural	mlp-train-step	gaussian	1.90	torch-compile-bf16 2.13.0+cu129	1.88	**1.01**×	loss_last_step 1.12	1.12
			1.90	torch-eager-bf16 2.13.0+cu129	1.80	**1.05**×		1.12
			1.90	torch-eager-fp32 2.13.0+cu129	1.83	**1.04**×		1.12
			1.90	torch-eager-tf32 2.13.0+cu129	1.68	**1.13**×		1.12
neural	ops-session-step	gaussian	2.30	torch-eager-tf32 2.13.0+cu129	2.10	**1.09**×	loss 2.80	2.80
neural	program-run	gaussian	1.04	torch-eager-bf16 2.13.0+cu129	0.694	**1.50**×	max_abs_err_vs_fp64 7.4e−8	4.91e−4
			1.04	torch-eager-fp32 2.13.0+cu129	0.655	**1.59**×		3.22e−8
			1.04	torch-eager-tf32 2.13.0+cu129	0.580	**1.79**×		3.73e−5
neural	samba-forward	bytes	4.60	torch-eager-bf16 2.13.0+cu129	4.33	**1.06**×	mean_nll 5.64	5.64
			4.60	torch-eager-fp32 2.13.0+cu129	3.98	**1.16**×		5.64
			4.60	torch-graph-fp32 2.13.0+cu129	1.72	**2.68**×		5.64

NVIDIA L40S, continued

family	algorithm	dataset	my ms	opponent	their ms	my ms ÷ their ms	my quality	their quality
			4.60	torch-graph-tf32 2.13.0+cu129	1.38	**3.34**×		5.64
neural	samba-train-step	bytes	45.7	torch-eager-bf16 2.13.0+cu129	19.3	**2.37**×	loss_last_step 4.82	4.82
			45.7	torch-eager-fp32 2.13.0+cu129	14.1	**3.25**×		4.82
			45.7	torch-eager-tf32 2.13.0+cu129	11.8	**3.88**×		4.82
neural	sequential-predict	synthetic	4.27	torch-compile-bf16 2.13.0+cu129	1.94	**2.20**×	auc 0.969	0.969
			4.27	torch-compile-fp32 2.13.0+cu129	1.61	**2.65**×		0.969
			4.27	torch-compile-tf32 2.13.0+cu129	2.65	**1.61**×		0.969
			4.27	torch-eager-bf16 2.13.0+cu129	1.49	**2.87**×		0.969
			4.27	torch-eager-fp32 2.13.0+cu129	1.76	**2.42**×		0.969
			4.27	torch-eager-tf32 2.13.0+cu129	1.38	**3.09**×		0.969
neural	transformer-decode	gaussian	1,100	torch-eager-bf16 2.13.0+cu129	862	**1.28**×	max_abs_diff_vs_host 0	–
			1,100	torch-eager-fp32 2.13.0+cu129	750	**1.47**×		–
			1,100	torch-eager-tf32 2.13.0+cu129	725	**1.52**×		–
			1,100	torch-graph-fp32 2.13.0+cu129	326	**3.38**×		–
			1,100	torch-graph-tf32 2.13.0+cu129	329	**3.36**×		–
neural	transformer-forward	gaussian	5.57	torch-compile-bf16 2.13.0+cu129	2.37	**2.35**×	rel_fro_vs_torch_eager_fp32 6.06e−8	2.63e−4
			5.57	torch-compile-fp32 2.13.0+cu129	2.68	**2.07**×		3.11e−8
			5.57	torch-compile-tf32 2.13.0+cu129	2.59	**2.15**×		2.66e−5
			5.57	torch-eager-bf16 2.13.0+cu129	3.66	**1.52**×		2.77e−4
			5.57	torch-eager-fp32 2.13.0+cu129	2.67	**2.09**×		–
			5.57	torch-eager-tf32 2.13.0+cu129	3.31	**1.68**×		2.66e−5
			5.57	torch-graph-fp32 2.13.0+cu129	2.36	**2.36**×		0
			5.57	torch-graph-tf32 2.13.0+cu129	3.31	**1.68**×		2.66e−5
classical	dbscan	istella	6.03e5	cuml-gpu 26.08.00	66,000	**9.13**×	n_clusters 40,100	40,100
classical	dbscan	taxi	1.58e6	cuml-gpu 26.08.00	4.4e5	**3.60**×	n_clusters 36.0	36.0
classical	hdbscan	istella	1,190	cuml-gpu 26.08.00	905	**1.32**×	silhouette −0.00386	−0.0182
classical	hdbscan	taxi	423	cuml-gpu 26.08.00	365	**1.16**×	silhouette 0.00150	0.0120
classical	svc (infer Xq)	istella	11.0	cuml-gpu 26.08.00§	4.90	**2.24**×	accuracy_eval 0.922	0.922
classical	svc	taxi	761	cuml-gpu 26.08.00	383	**1.98**×	accuracy 0.767	0.767
classical	svc (infer Xq)	taxi	6.07	cuml-gpu 26.08.00§	4.65	**1.31**×	accuracy_eval 0.767	0.767
classical2	kernel-ridge	istella	533	cuml-gpu 26.08.00	341	**1.56**×	r2 0.385	0.385
classical2	kernel-ridge	taxi	533	cuml-gpu 26.08.00	337	**1.58**×	r2 0.727	0.727
classical2	linearsvc	taxi	68.7	cuml-gpu 26.08.00	61.0	**1.13**×	accuracy 0.763	0.763
classical2	svr	istella	63.9	cuml-gpu 26.08.00	58.8	**1.09**×	r2 0.318	0.318
classical2	svr	taxi	65.3	cuml-gpu 26.08.00	49.1	**1.33**×	r2 0.768	0.768
trees	gbdt-depthwise (infer large)	istella	not-scored	catboost-gpu 1.2.10	not-scored	–	–	–
trees	gbdt-depthwise (infer large)	taxi	not-scored	catboost-gpu 1.2.10	not-scored	–	–	–

NVIDIA L40S, continued

family	algorithm	dataset	my ms	opponent	their ms	my ms ÷ their ms	my quality	their quality
trees	gbdt-lossguide	taxi	64,300	catboost-gpu 1.2.10	20,500	**3.14**×	auc 0.632	0.632
			63,000	xgboost-gpu 3.2.0	11,400	**5.52**×		0.632
trees	gbdt-lossguide (infer large)	taxi	not-scored	catboost-gpu 1.2.10	not-scored	–	–	–
			not-scored	xgboost-gpu 3.2.0	not-scored	–		–
trees	gbdt-lossguide (infer test)	taxi	9.88	xgboost-gpu 3.2.0	7.91	**1.25**×	auc 0.632	0.632
trees	gbdt-multiclass (infer large)	istella	not-scored	catboost-gpu 1.2.10	not-scored	–	–	–
trees	gbdt-multiclass (infer large)	taxi	not-scored	catboost-gpu 1.2.10	not-scored	–	–	–
trees	gbdt-rank-pairlogit (infer large)	istella	not-scored	catboost-gpu 1.2.10	not-scored	–	–	–
trees	gbdt-symmetric	istella	9,950	catboost-gpu 1.2.10	7,450	**1.34**×	auc 0.980	0.980
trees	gbdt-symmetric (infer large)	istella	not-scored	catboost-gpu 1.2.10	not-scored	–	–	–
trees	gbdt-symmetric	taxi	6,150	catboost-gpu 1.2.10	5,480	**1.12**×	auc 0.630	0.630
trees	gbdt-symmetric (infer large)	taxi	not-scored	catboost-gpu 1.2.10	not-scored	–	–	–
trees	iforest (infer large)	istella	not-scored	cuml-iforest-gpu 26.08.00	not-scored	–	–	–
trees	iforest	taxi	149	cuml-iforest-gpu 26.08.00	114	**1.31**×	auc 0.547	0.552
trees	iforest (infer large)	taxi	not-scored	cuml-iforest-gpu 26.08.00	not-scored	–	–	–
trees	rf	istella	30,800	cuml-rf-gpu 26.08.00	23,700	**1.30**×	auc 0.945	0.945
trees	rf (infer large)	istella	not-scored	cuml-rf-gpu 26.08.00	not-scored	–	–	–
trees	rf (infer test)	istella	52.3	cuml-rf-gpu 26.08.00	47.8	**1.09**×	auc 0.945	0.945
trees	rf (infer large)	taxi	not-scored	cuml-rf-gpu 26.08.00	not-scored	–	–	–

2 AMD MI300X

Table 2: AMD MI300X: fit and inference comparisons. Times in milliseconds; ratios divide my runtime by the opponent's. Quality is shown alongside runtime. Snapshot: October 30, 2026.

family	algorithm	dataset	my ms	opponent	their ms	my ms ÷ their ms	my quality	their quality
neural	gemm	gaussian	65.8	torch-compile-bf16 2.13.0+rocm7.1	20.2	**3.26**×	max_rel_err_vs_fp64 2.4e−7	0.00375
			65.8	torch-compile-fp32 2.13.0+rocm7.1	22.3	**2.95**×		2.8e−6
			65.8	torch-eager-bf16 2.13.0+rocm7.1	19.3	**3.41**×		0.00375
			65.8	torch-eager-fp32 2.13.0+rocm7.1	20.4	**3.23**×		2.8e−6
neural	gemm-bf16	gaussian	48.5	torch-compile-bf16 2.13.0+rocm7.1	18.3	**2.65**×	max_rel_err_vs_fp64 1.15e−7	0.00276
			48.5	torch-eager-bf16 2.13.0+rocm7.1	17.1	**2.83**×		0.00276
neural	lm-forward	bytes	58.5	torch-compile-bf16 2.13.0+rocm7.1	20.4	**2.86**×	mean_nll 9.02	9.02
			58.5	torch-compile-fp32 2.13.0+rocm7.1	21.1	**2.77**×		9.02
			58.5	torch-eager-bf16 2.13.0+rocm7.1	25.3	**2.31**×		9.02
			58.5	torch-eager-fp32 2.13.0+rocm7.1	18.4	**3.18**×		9.02

AMD MI300X, continued

family	algorithm	dataset	my ms	opponent	their ms	my ms ÷ their ms	my quality	their quality
			58.5	torch-graph-fp32 2.13.0+rocm7.1	21.3	**2.75**×		9.02
neural	lm-train-step	bytes	43.9	torch-compile-bf16 2.13.0+rocm7.1	9.70	**4.53**×	loss_last_step 8.41	8.40
			43.9	torch-compile-fp32 2.13.0+rocm7.1	14.7	**2.99**×		8.41
			43.9	torch-eager-bf16 2.13.0+rocm7.1	19.4	**2.26**×		8.40
			43.9	torch-eager-fp32 2.13.0+rocm7.1	17.0	**2.58**×		8.41
neural	mamba1-forward	gaussian	3.02	mamba-ssm-fp32 2.3.2.post1+ge9594ce1c732	1.40	**2.15**×	rel_fro_vs_torch_eager_fp32 7.44e−9	7.34e−9
neural	mamba2-forward	gaussian	5.28	mamba-ssm-fp32 2.3.2.post1+ge9594ce1c732	2.71	**1.95**×	rel_fro_vs_torch_eager_fp32 1.37e−7	7.52e−8
neural	mlp-train-step	gaussian	3.32	torch-compile-bf16 2.13.0+rocm7.1	1.80	**1.84**×	loss_last_step 1.12	1.12
			3.32	torch-compile-fp32 2.13.0+rocm7.1	2.07	**1.60**×		1.12
			3.32	torch-eager-bf16 2.13.0+rocm7.1	1.90	**1.74**×		1.12
			3.32	torch-eager-fp32 2.13.0+rocm7.1	1.85	**1.80**×		1.12
neural	ops-session-step	gaussian	2.39	torch-compile-bf16 2.13.0+rocm7.1	2.22	**1.08**×	loss 2.80	2.80
neural	program-run	gaussian	1.25	torch-compile-bf16 2.13.0+rocm7.1	0.663	**1.88**×	max_abs_err_vs_fp64 7.4e−8	4.91e−4
			1.25	torch-compile-fp32 2.13.0+rocm7.1	0.918	**1.36**×		3.38e−8
			1.25	torch-eager-bf16 2.13.0+rocm7.1	0.675	**1.85**×		4.91e−4
			1.25	torch-eager-fp32 2.13.0+rocm7.1	0.594	**2.10**×		3.38e−8
neural	sequential-predict	synthetic	4.22	torch-compile-bf16 2.13.0+rocm7.1	1.45	**2.90**×	auc 0.969	0.969
			4.22	torch-compile-fp32 2.13.0+rocm7.1	1.47	**2.88**×		0.969
			4.22	torch-eager-bf16 2.13.0+rocm7.1	1.24	**3.41**×		0.969
			4.22	torch-eager-fp32 2.13.0+rocm7.1	1.13	**3.73**×		0.969
neural	transformer-decode	gaussian	2,440	torch-eager-bf16 2.13.0+rocm7.1	1,090	**2.23**×	max_abs_diff_vs_host 0	–
			2,440	torch-eager-fp32 2.13.0+rocm7.1	929	**2.63**×		–
neural	transformer-forward	gaussian	8.04	torch-compile-bf16 2.13.0+rocm7.1	2.20	**3.66**×	rel_fro_vs_torch_eager_fp32 5.72e−8	2.63e−4
			8.04	torch-compile-fp32 2.13.0+rocm7.1	1.41	**5.70**×		3.1e−8
			8.04	torch-eager-bf16 2.13.0+rocm7.1	1.71	**4.71**×		2.76e−4
			8.04	torch-eager-fp32 2.13.0+rocm7.1	1.59	**5.05**×		–
			8.04	torch-graph-fp32 2.13.0+rocm7.1	0.930	**8.64**×		0
trees	gbdt-depthwise	taxi	3,550	xgboost-gpu 3.2.0	3,260	**1.09**×	auc 0.632	0.632
trees	gbdt-lossguide	taxi	19,500	xgboost-gpu 3.2.0	17,700	**1.10**×	auc 0.632	0.632
trees	gbdt-lossguide (infer large)	taxi	not-scored	xgboost-gpu 3.2.0	not-scored	–	–	–